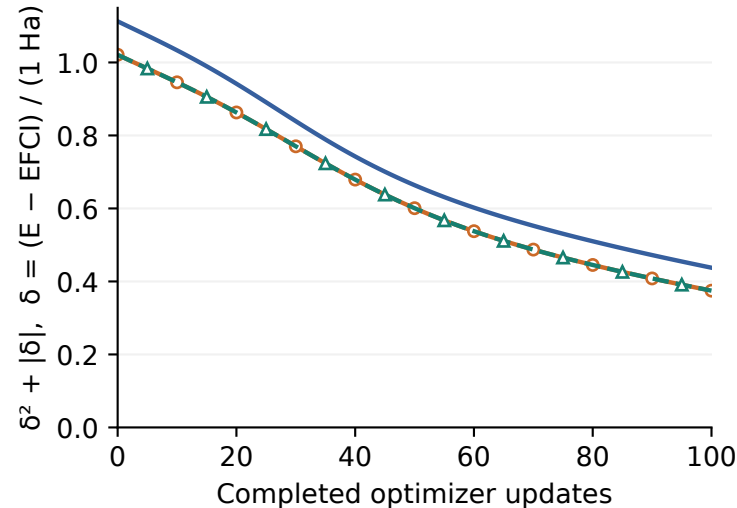


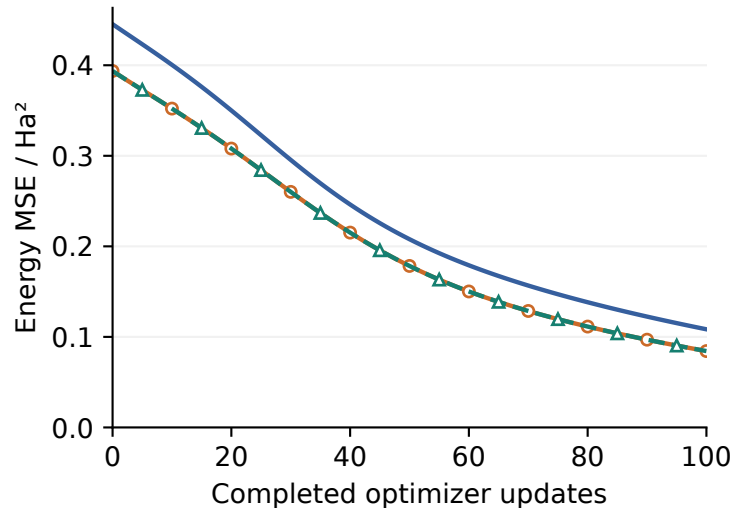
H₂ / 6-31G* · Four-layer density-matrix MLP · FCI supervision

— Fixed HF density —○— Self-consistent / unrolled —△— Self-consistent / implicit

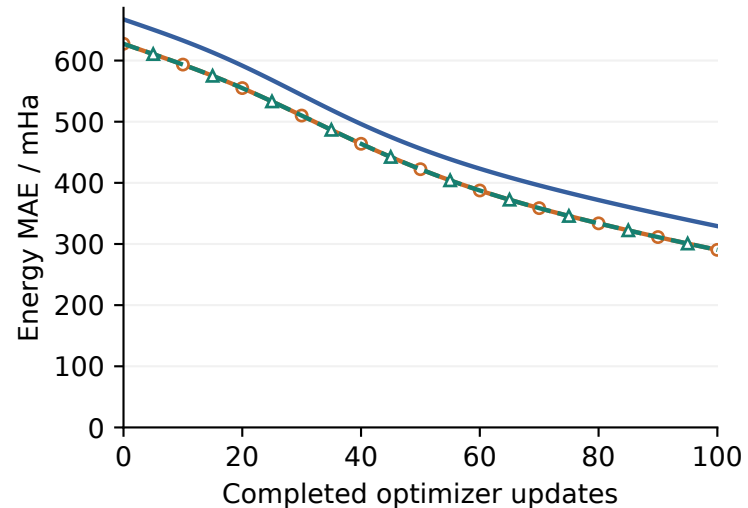
Total loss



MSE component



MAE component



E_FCI = -1.151672544961 Ha. Adam lr=0.002; seed=0; CPU float64; 100/100 updates accepted per mode.

Unrolled and implicit curves overlap. Fixed-density loss uses the unchanged HF state; the other curves use converged SCF states.